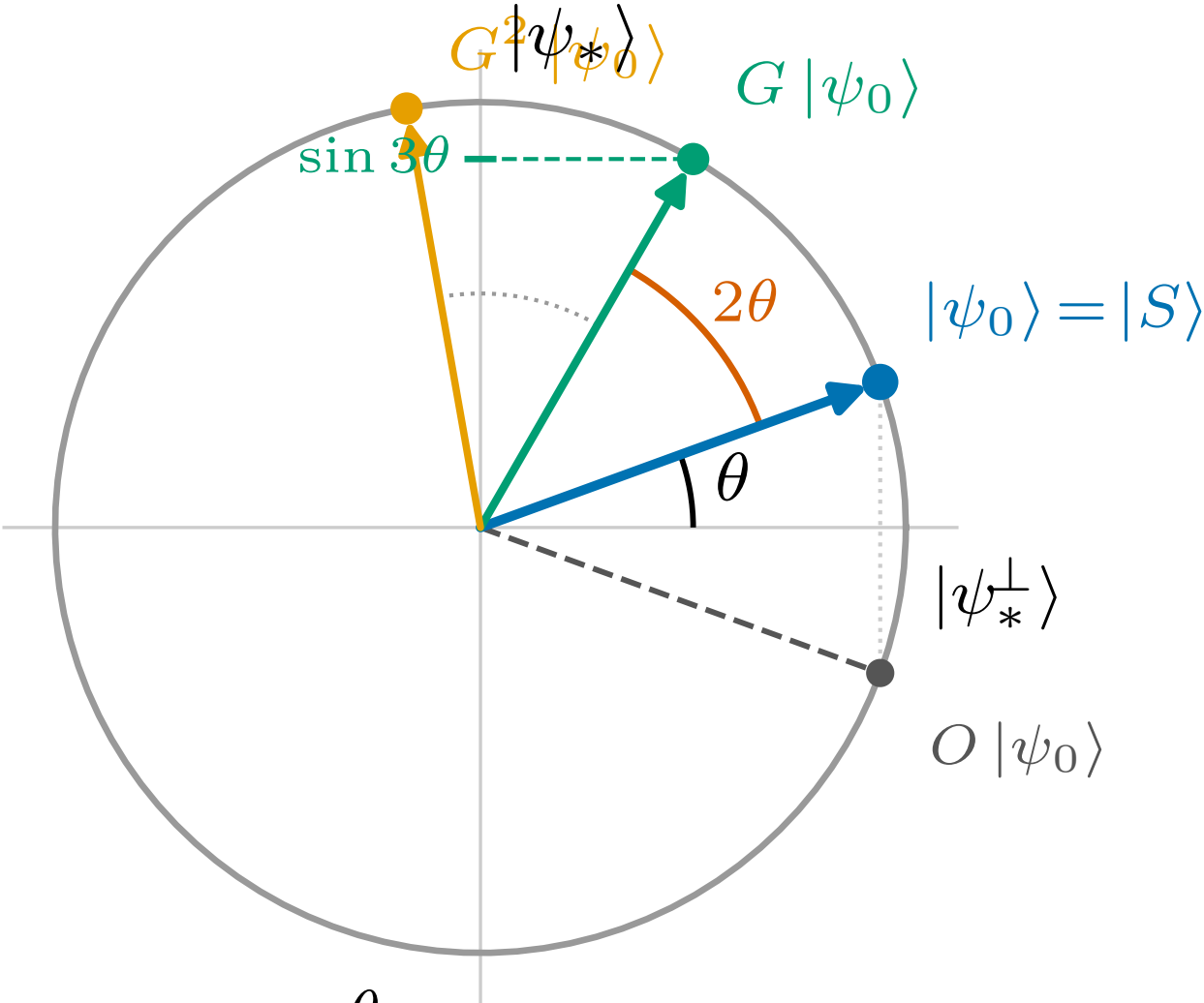
$|\psi_*\rangle$
$G^2|\psi_0\rangle$
$G|\psi_0\rangle$
$\sin 3\theta$
2θ
$|\psi_0\rangle = |S\rangle$
θ
$|\psi_*^\perp\rangle$
$O|\psi_0\rangle$

$a = \cos\theta$